QC · Ethylene · inZOR-ND · 3D

Ethylene (3D) — Quasi-Degenerate Gap Structure via inZOR-ND

- **Discovery speed** is evaluated using a threshold of **gap < 10 meV** (first-hit probe index; engine vs uniform control in the table below).
- **Cluster analysis** is performed on **all points with gap < 0.01 eV** (pooled across seeds for DBSCAN and figures) — several quasi-degenerate basins, not a single global funnel.
- **What is shown: not** "one correct region found" — rather, **multiple** quasi-degenerate basins in geometry space, with discovery speed and Top-K emphasis that **depend on seed**.
- **Cross-seed:** several low-gap regions recur, but **dominance varies by seed** (e.g. Top-K=50 weight in sector B ≈ 0.42 for seed 42, ≈ 0.08 for 137, ≈ 0.30 for 2011).
- **Determinism (same seed):** fixed seed + fixed protocol → byte-identical `probes.jsonl` on rerun — **not** a claim that different seeds or different trajectories coincide (see reproducibility).

Setup

Parameter	Value
Dimensions	3D (torsion, r_{CC}, z_lift)
Method	SA-CASSCF(2,2)
Seeds	42, 137, 2011
Steps	15
Population	12 → 30

Score variant `legacy`. Main tables: regional batch `validation_regional_3seeds_20260403_155239`. Additional full regional run (seed 42 only): `validation_20260403_211825_legacy_nd3_regional_s42` — its `zor_seed_42/probes.jsonl` is **byte-identical** to seed 42 in the 3-seed batch (`cmp` exit 0).

What this demonstrates

> The evidence here is **not** that inZOR-ND identifies a single privileged "right" region in internal coordinates. It **is** that, in 3D, the run visits and ranks **several** separated low-gap regions (clusters, sector weights, pooled scatter) — i.e. **multi-basin quasi-degeneracy**, not funnel collapse to one geometry.
>
> **Best gap** (smallest S_0–S_1 seen in a run) is one scalar summary per seed. **Landscape structure** — where probes pile up, how many DBSCAN basins appear, whether the argmin sits in a supported patch — is a separate object: you need both, but the global minimum alone does not describe the geometry-space organisation this page is about.

Main results (per seed)

Seed	Best gap (eV)	First <10 meV (probe #)	Top-K dominant region (K=50)
42	$\sim 2.6\times10^{-4}$	4	B (~260°; circular mean ~238°)
137	$\sim 2.0\times10^{-4}$	87	A (~70–90°)
2011	$\sim 7.3\times10^{-4}$	316	B (~240–260°)

> **Global best-gap is not an isolated spike.** In `validation_regional_3seeds_summary.json`, `best_appears_isolated` is **false** for seeds 42, 137, and 2011: the minimum-gap geometry has real local support in the patch metric. The **local 5×5 grid** and **z_{lift} sweep** figures below visualize that coherence around the best point.

inZOR-ND vs uniform random (matched budget)

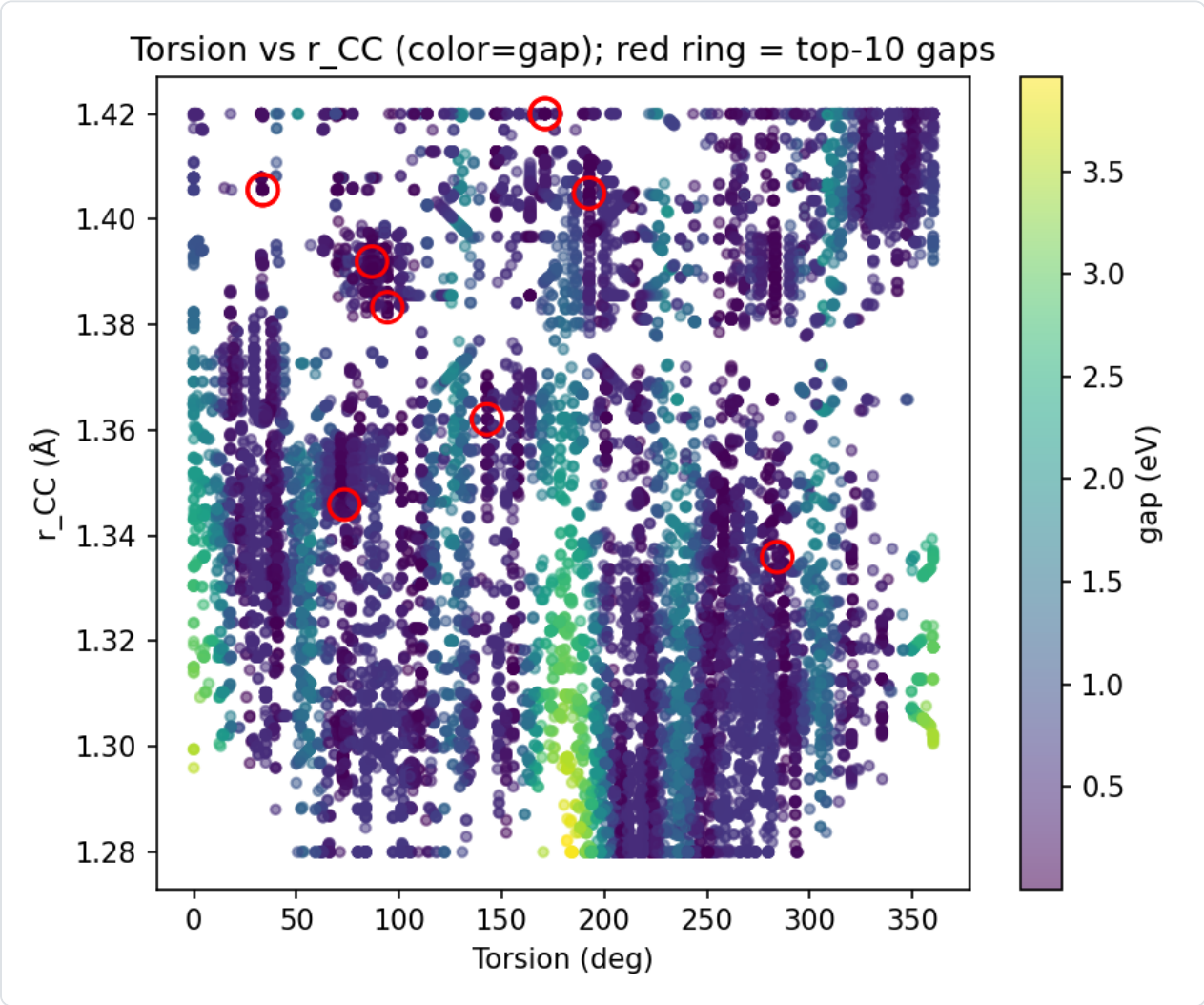
Metric	inZOR-ND	Random
Best gap	$\sim 2.0\times10^{-4}$ eV (best over seeds)	$\sim 5.7\times10^{-4}$ eV
First <10 meV	4 (fastest seed: 42)	141

Control: n_eval = 8768 uniform samples in $[0,1]^3$ (same internal-coordinate mapping as this 3D engine). inZOR-ND reaches a sub-10 meV gap far earlier than uniform sampling on this budget (e.g. first hit at probe 4 vs 141 in this batch).

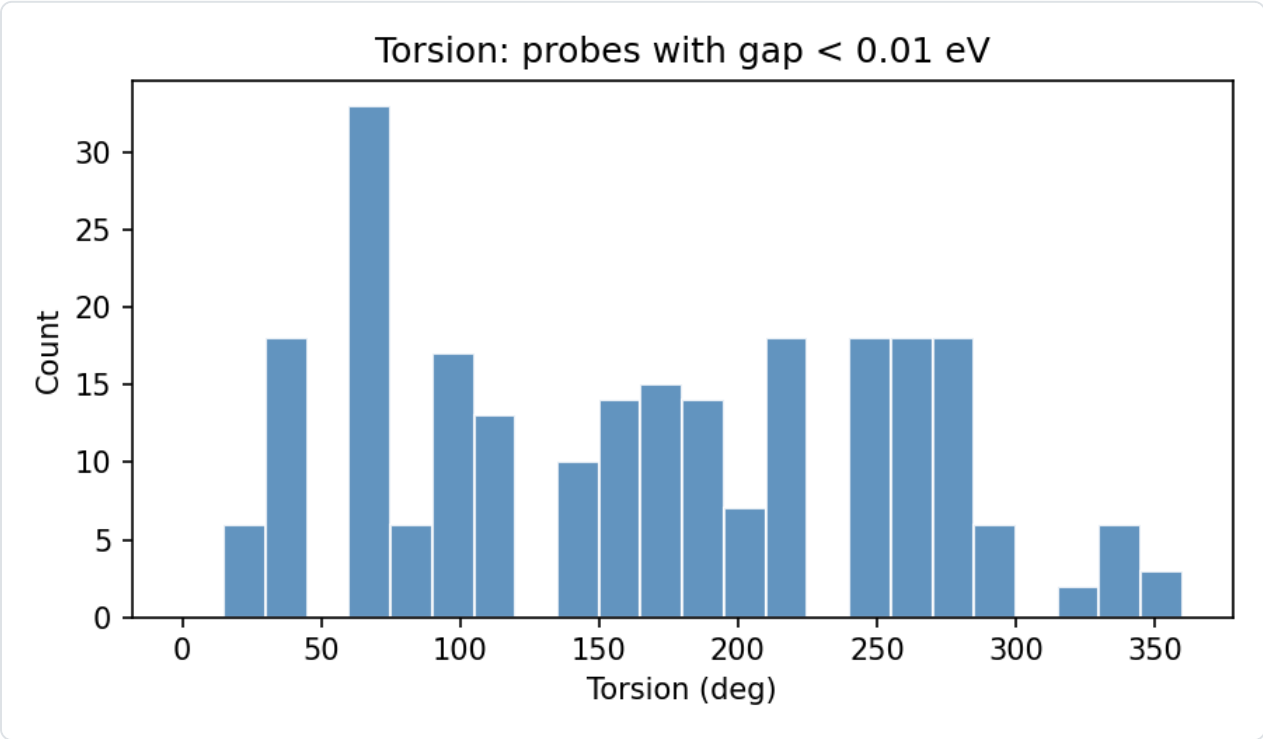
Figures (this 3D report's validation run)

These PNGs are emitted by `validate_ethylene_zor.py` into `logs/validation_regional_3seeds_20260403_155239/` (3 seeds, 3D regional mode) — **not** ad-hoc graphics from unrelated folders.

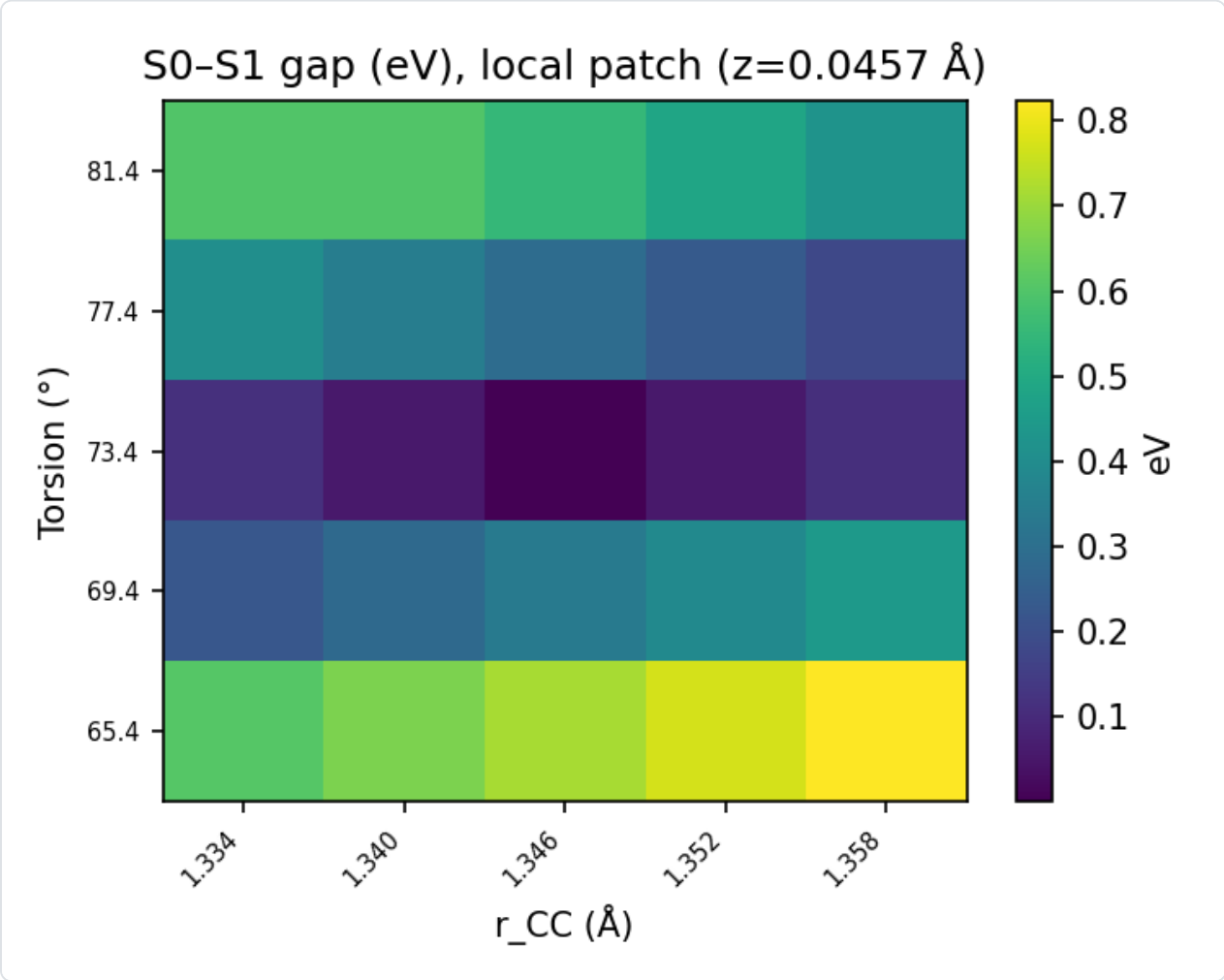
Pooled vs per-seed: the scatter, low-gap histogram, local grid, and z sweep are built from **concatenated probes** across seeds (same logic as regional sections H–J in the full validation write-up). **Per-seed** regional metrics (sector weights, first <10 meV, isolation flags, etc.) live in `validation_regional_3seeds_summary.json` — summarized in the main table above, not re-plotted here line-by-line.



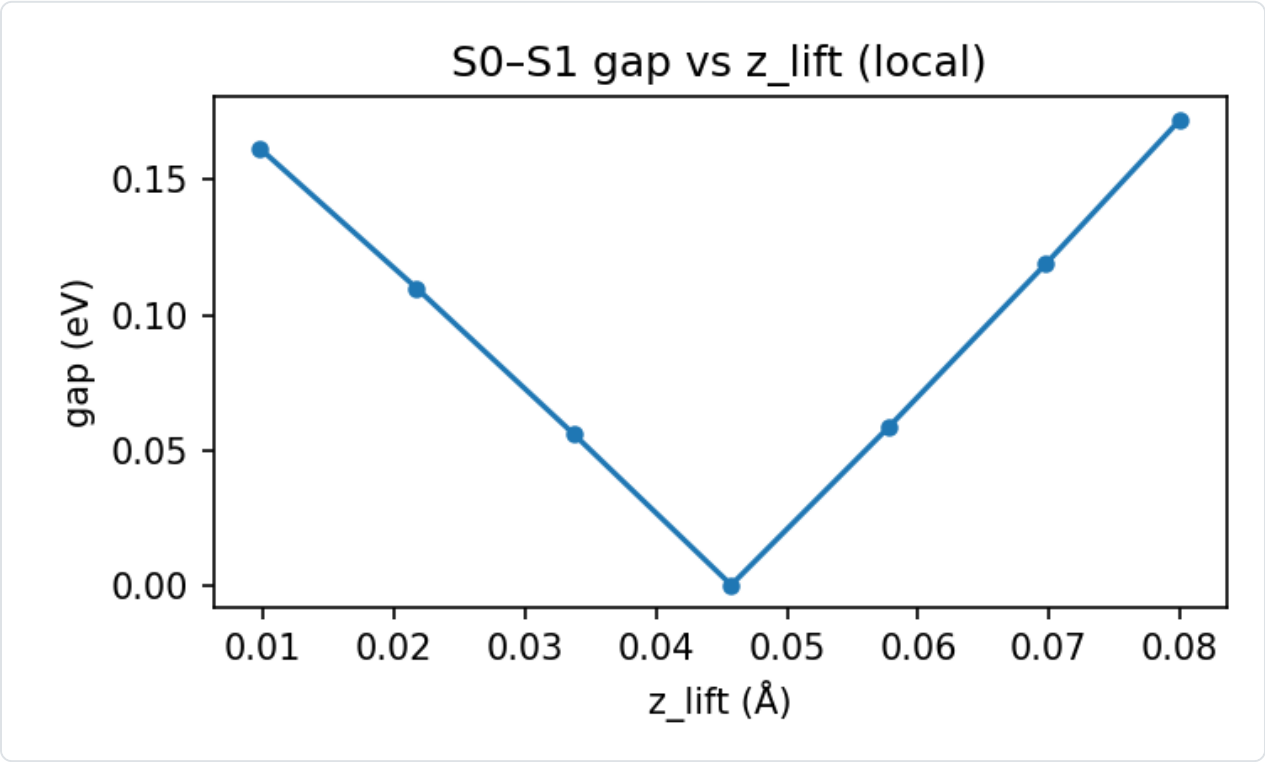
Scatter (torsion, r_{CC}) coloured by S_0–S_1 gap — pooled / regional view from that validation folder.



Low-gap torsion histogram — concentration of quasi-degenerate probes in angle.



Local 5×5 gap grid in (torsion, r_{CC}) at the best point's z_{lift} (coherence check).



z_{lift} sweep — gap profile out-of-plane around the best geometry.

Cluster analysis (gap < 0.01 eV, all seeds pooled)

DBSCAN on (cos θ, sin θ, r_{CC}, z_{lift}); hyperparameters match the regional validation cluster export used for this report.

- Total low-gap points: 242
- Total clusters: 15
- Largest cluster: ~11% of non-noise points (16 / 149)
- Substantial clusters (rule in JSON): 6
- Multi-seed clusters: 6

Cluster	Torsion (°)	Size	Multi-seed
C1	~70	14	yes
C2	~171	15	yes
C3	~258	12	yes
C4	~91	16	no
C5	~252	9	yes
C6	~145	8	no

Rows ordered by scientific interest; sizes and angles from exported centroids.

Pooled vs per-seed: Cluster analysis is performed on pooled probes across all seeds, while regional consistency metrics are evaluated per seed.

Interpretation

> **Several regions recur across seeds, but their dominance varies by seed.** The exploration does not collapse to a single minimum: multiple quasi-degenerate regions appear in the pooled low-gap set. Recurrence is structured multi-basin behaviour, not uniform repetition across seeds (see Top-K sector weights in the main table and regional summary). The main scientific point is this **regional layout** of the gap in geometry space — not only how low the best gap happens to be on a given seed.

Regional consistency

- 2 of 3 seeds put more Top-K weight in sector B than A; the third seed (137) concentrates differently — real heterogeneity between runs.
- Concrete Top-K=50 sector-B fractions from the regional summary: ≈0.42 (seed 42), ≈0.08 (seed 137), ≈0.30 (seed 2011).
- In all seeds, the best-gap configuration is not isolated and has local support in the surrounding region.
- All seeds: `best_appears_isolated: false` — the best-gap geometry sits in a supported local patch (see figures + JSON).
- Low-gap and top-K distributions differ by seed, but concentration in chemically plausible torsion bands persists.

Reproducibility & cluster stability (latest technical step)

> **If you skim only one line:** this section shows **determinism when the seed and protocol are fixed** (rerun → same `probes.jsonl`). It is **not** evidence that the algorithm would match itself across different seeds or across two independent trajectories that actually diverge at the file level.

> **Reproducibility is demonstrated for identical runs** (same seed and protocol), which produce **identical probe trajectories** and **identical clusters**.
>
> **This does not imply stability** across independent runs that generate different probe trajectories. The **HIGH** verdicts in the table apply only when `cmp` on `probes.jsonl` is identical (byte-for-byte).

Scripts (under `scene_ethylene_quasidegen/` in the source tree): `low_gap_cluster_analysis.py` — `cluster_probes_jsonl(...)` for one `probes.jsonl` (no multi-seed aggregation). `compare_low_gap_cluster_stability.py` — compares run A vs B (`--dir-a` / `--dir-b` or explicit paths), re-clusters low-gap points with the same DBSCAN feature space (cos/sin torsion, r_{CC}, z_{lift}), scales on the union of low-gap points, then reports match rate, centroid distances, approximate overlap, and verdict **high** / moderate / low.

Comparison (seed 42)	`cmp` on `probes.jsonl`	Low-gap (<0.01 eV)	Clusters A/B	Matched	Mean centroid Δ (scaled)	V
`validation_20260403_130511…_regional_s42` vs `…155239/zor_seed_42`	identical (exit 0)	92 / 92	7 / 7	7/7	0	H
`…155239/zor_seed_42` vs `validation_20260403_211825…_regional_s42`	identical (exit 0)	92 / 92	7 / 7	7/7	0	H

Machine-readable output for the second row is mirrored here: `ethylene_3d_artifacts/cluster_stability_rerun_211825_vs_155239.json` and `ethylene_3d_artifacts/CLUSTER_STABILITY_RERUN_211825_vs_155239.txt`.

> **Cross-seed structure:** use the pooled cluster table + per-seed metrics above (heterogeneity is expected).
> **Same-run reproducibility:** identical `probes.jsonl` → identical low-gap DBSCAN clusters here (HIGH, zero centroid drift) — validates determinism + clustering code path for that trace only.
> **Still open:** cluster stability for two independent runs that produce different probe streams under the same protocol.

Limitations

- Global best-gap geometry is not guaranteed to fall inside fixed A/B sectors.
- Dominant top-K region varies between seeds.
- Clusters are multiple basins, not a single basin.

Conclusion

> This result should **not** be interpreted as convergence to a single optimal geometry.

> Instead, the system exhibits **multiple quasi-degenerate regions** in geometry space. inZOR-ND exploration does **not** select a unique minimum, but **efficiently discovers several low-gap regions**, some of which recur across seeds.
>
> The global minimum alone is therefore **not sufficient** to describe the structure of the problem.
>
> In short: inZOR-ND in this 3D setting reveals a **structured family** of quasi-degenerate configurations rather than collapse to one basin.

← QC gaps: unified report (H_2 & ethylene) · Tests catalogue